

$$\frac{H_1}{H_2} \leq G \quad H_1 H_2 \leq G \Leftrightarrow H_1 H_2 = H_2 H_1.$$

Cor $N \trianglelefteq G \quad H \leq G \Rightarrow {}^G H N = N H \leq G.$

Pf: $\forall \alpha \in H \Rightarrow \alpha N = N \alpha \Leftrightarrow \alpha N \alpha^{-1} = N.$
 $\forall \alpha \in \alpha N \Rightarrow \alpha \alpha^{-1} = \underbrace{\alpha \alpha^{-1}}_{\in N} \alpha \in N \alpha$

$$H N = \bigcup_{\alpha \in H} \alpha N = \bigcup_{\alpha \in H} N \alpha = N H$$

② $N \trianglelefteq H N \quad (N \trianglelefteq G) \quad H N \leq G.$

③ $H \cap N \trianglelefteq H.$

$$\forall \alpha \in H \cap N \quad \forall h \in H.$$

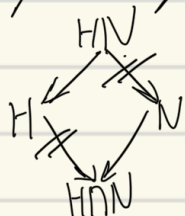
$$h \alpha h^{-1} \in H \quad (\because \alpha, h \in H)$$

Since $\alpha \in N$ $h \alpha h^{-1} \in N.$

$$\Rightarrow h \alpha h^{-1} \in H \cap N$$

Thm. $N \trianglelefteq G. \quad H \leq G.$

$$H N / N \cong H / H \cap N.$$



Pf: $\varphi: H \rightarrow H N / N$
 $h \mapsto h N$

$$\varphi(h_1 h_2) = h_1 h_2 N = (h_1 N)(h_2 N) = \varphi(h_1) \varphi(h_2)$$

φ is epic

$$\text{Ker } \varphi = \{h \in H / h N = N\}$$

$$= H \cap N$$

$$H N / N \cong H / H \cap N.$$

$$\dim V_1 \cap V_2 / V_2 = \dim V_1 / U \cap V_2$$

$$\dim(V_1 \cap V_2) - \dim V_2 = \dim V_1 - \dim(U \cap V_2).$$

没有推数. 只好把同构推到前边

作业. B7. 4. 9. 12. 16. 18. 21.